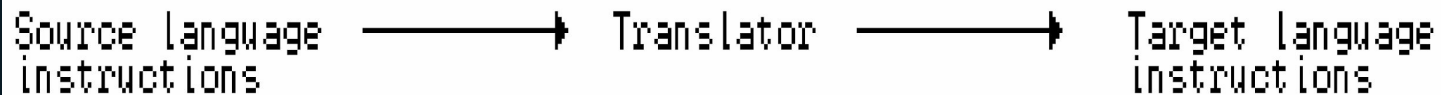


# Bootstrapping

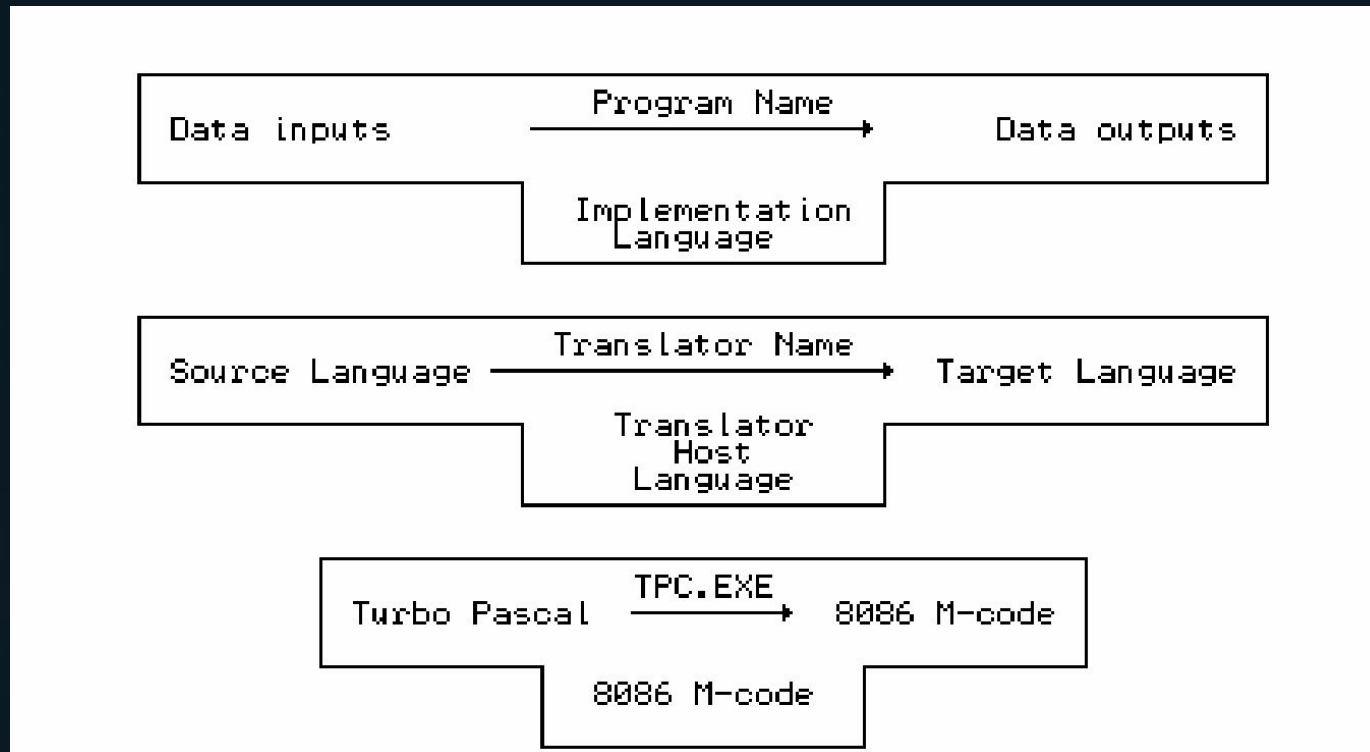
# Introduction

- At least three Languages are involved in the development of translators: the source language to be translated, the object or target language to be generated, and the host language to be used for implementing the translator.
- If the translation takes place in several stages, there may even be other, intermediate, languages.



# T-diagrams

- A useful notation for describing a computer program, particularly a translator called as T-diagrams.

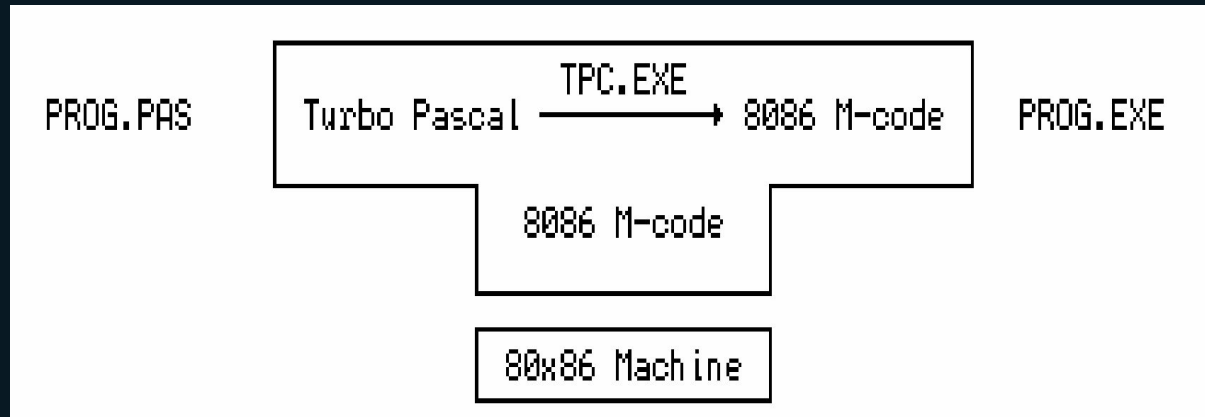


M-code:  
machine  
code

1. A General Program
2. A General Translator
3. A Turbo Pascal Compiler For an MS-DOS System

- Source: Compilers and Compiler Generators an introduction with C++ by P.D. Terry, Rhodes University.

# T-diagrams Cont..

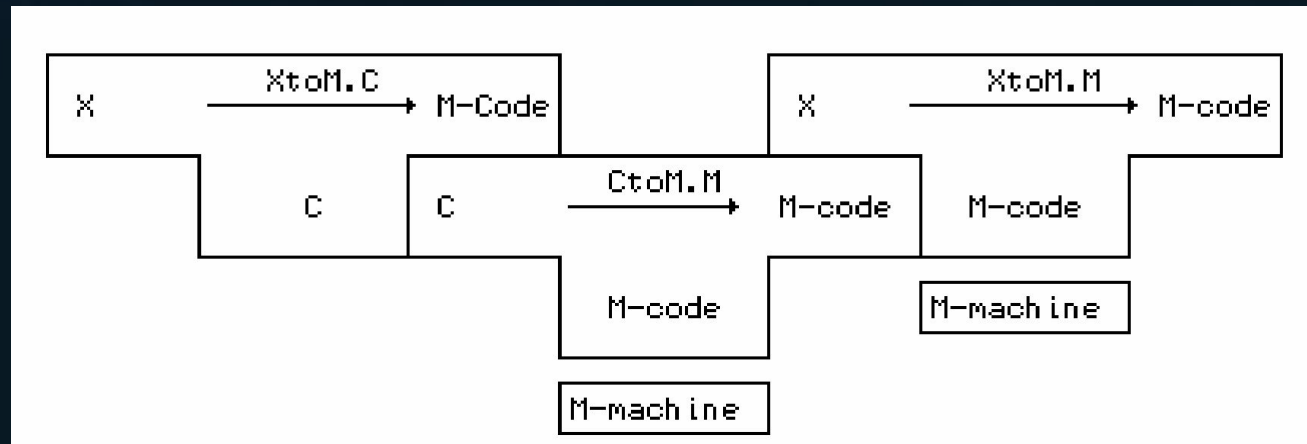


## A Turbo Pascal Compilation

An abstract machine (sometimes called a virtual machine), to convert Turbo Pascal source programs into their 8086 machine code equivalents

- Source: Compilers and Compiler Generators an introduction with C++ by P.D. Terry, Rhodes University.

# Using a high-level host language



## Use of C as an Implementation Language

- We have been given a machine M with the machine coded version of a compiler for a well-established language like C and to generate a compiler for a new language X.
- We need to write the new compiler, say XtoM, in C and compile the source (XtoM.C) with the C compiler (CtoM.M) running directly on M.
- We get the object version (XtoM.M) which can then be executed on M.

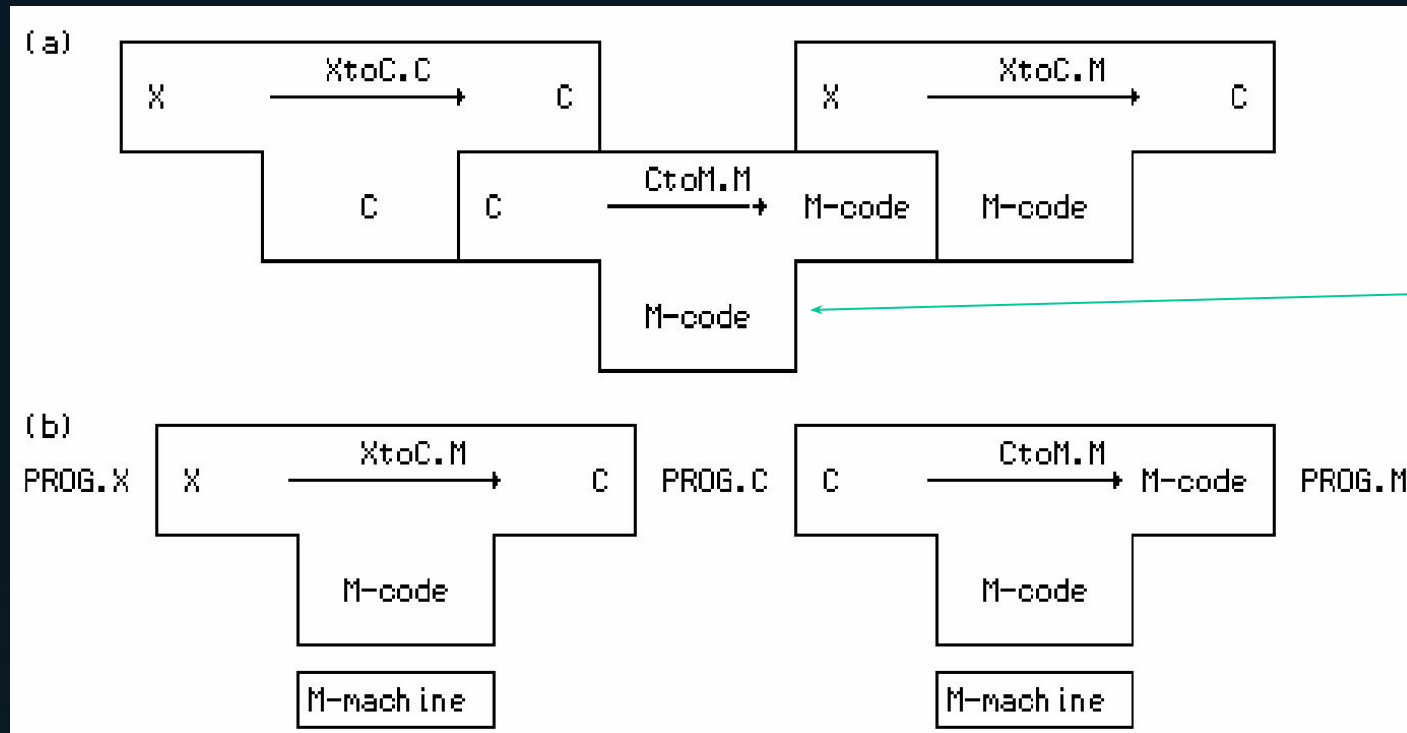
# Using a high-level host language

- Development in C is going to be easier than development in machine code
- This process is still complex.
- Need to develop a large part of the compiler source using compiler generator tools, namely Flex (Lexical Analysis) and YACC-Yet Another Compiler Compiler (Syntax Analysis)
- These tools should be available in executable form, or as C source that can itself be compiled.

# Porting a high level translator

- The process of modifying an existing compiler to work on a new machine is called as porting the compiler.
- A compiler XtoC for a language X has been implemented in C on machine A by writing a high-level translator to convert programs written in X to C
- We want to use language X on a machine M that, like A, has a C compiler of its own.
- We want to construct a two-stage compiler for use on either machine (i.e. M or A), then we need to install the source code for XtoC on machine M and recompile it.

# Porting a high level translator



On  
Machine A

Porting and Using a High-level Translator